Computer Networks

Lecture 8: Data Link Layer

Dr. Tauheed Ahmed

Content

- **Error Detection and Correction**

Data Link Layer

The data link layer transforms the physical layer, a raw transmission facility, to a link responsible for node-to-node (hop-to-hop) communication.

Specific responsibilities of the data link layer include

- framing,
- addressing,
- flow control,
- error control, and
- media access control.

Error Detection and Correction

Data can be corrupted during transmission.

Some applications can tolerate a small level of error.

For example:
Random errors in audio or video transmissions may be tolerable, but when we transfer text, we expect a very high level of accuracy.

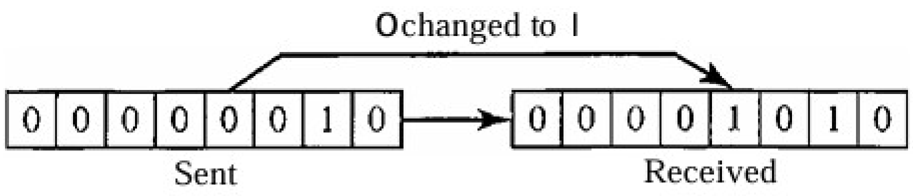
Some applications require that errors be detected and corrected.

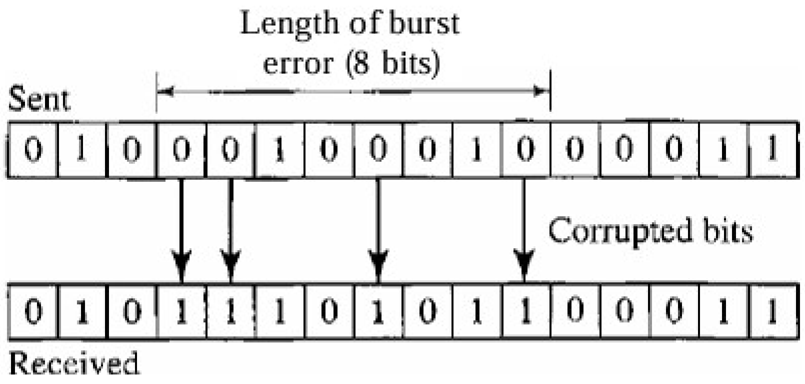
For example:
when we transfer text, we expect a very high level ofaccuracy

Types of Error

Whenever bits flow from one point to another, they are subject to unpredictable changes because of interference. This interference can cause

- Single-Bit Error
- Burst Error





Redundancy

The central concept in detecting or correcting errors is redundancy. To be able to detect or correct errors, we need to send some extra bits with our data.

Error Detection: In error detection, we are looking only to see if any error has occurred. The answer is a simple ***yes or no***.

Error Correction: In error correction, we need to know the exact number of bits that are corrupted and more importantly, their location in the message.

Error Correction

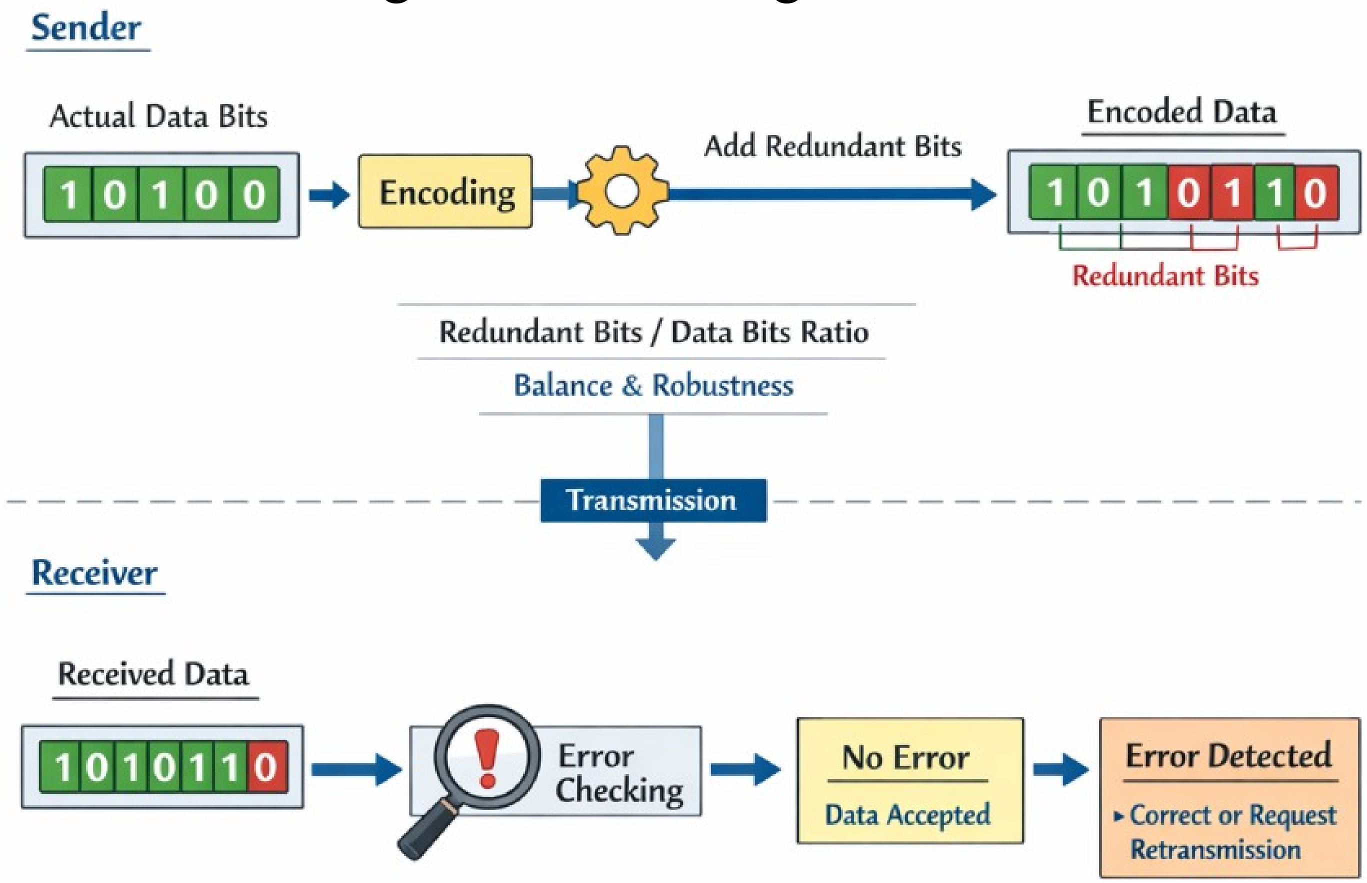
Approach There are two main methods of error correction.

Forward error correction is the process in which the receiver tries to guess the message by using redundant bits. This is possible, as we see later, if the number of errors is small.

Correction by retransmission is a technique in which the receiver detects the occurrence of an error and asks the sender to resend the message.

Coding

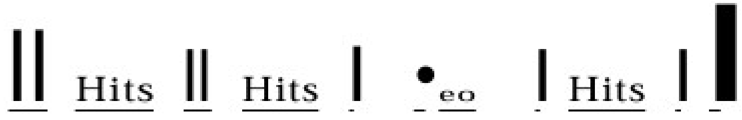
Redundancy is achieved through various coding schemes.



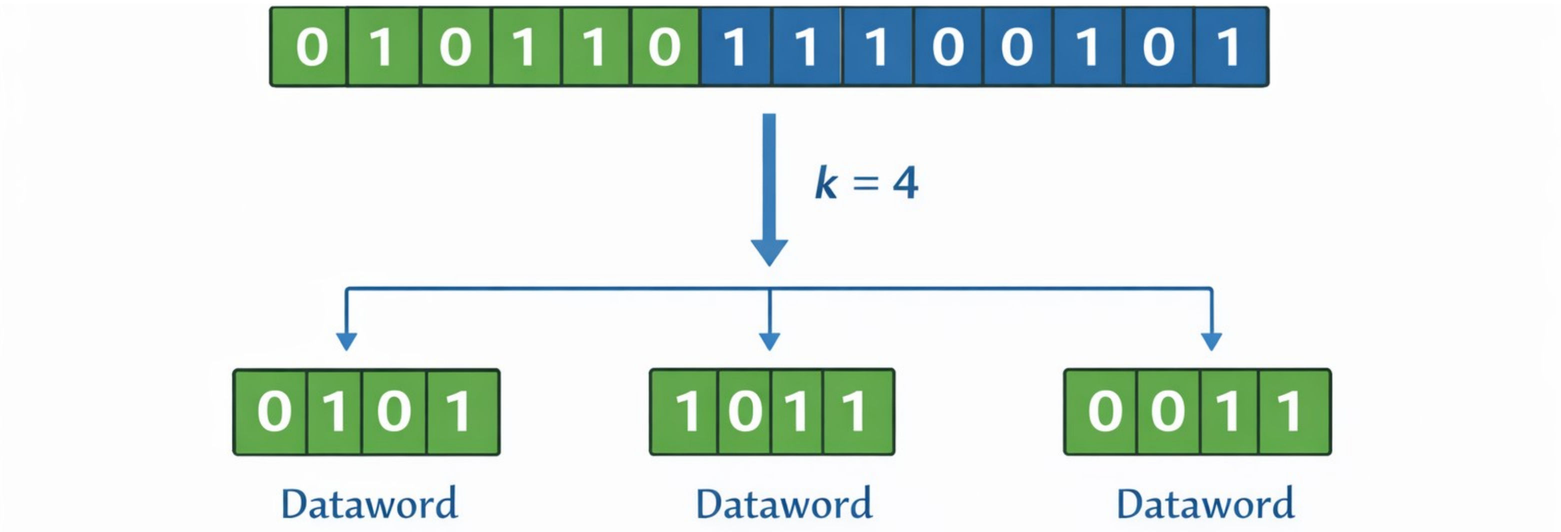
Block Coding

Block Coding is an error-control technique in which a block of data bits is replaced by a larger block of coded bits

We divide our message into blocks, each of k bits, called datawords.



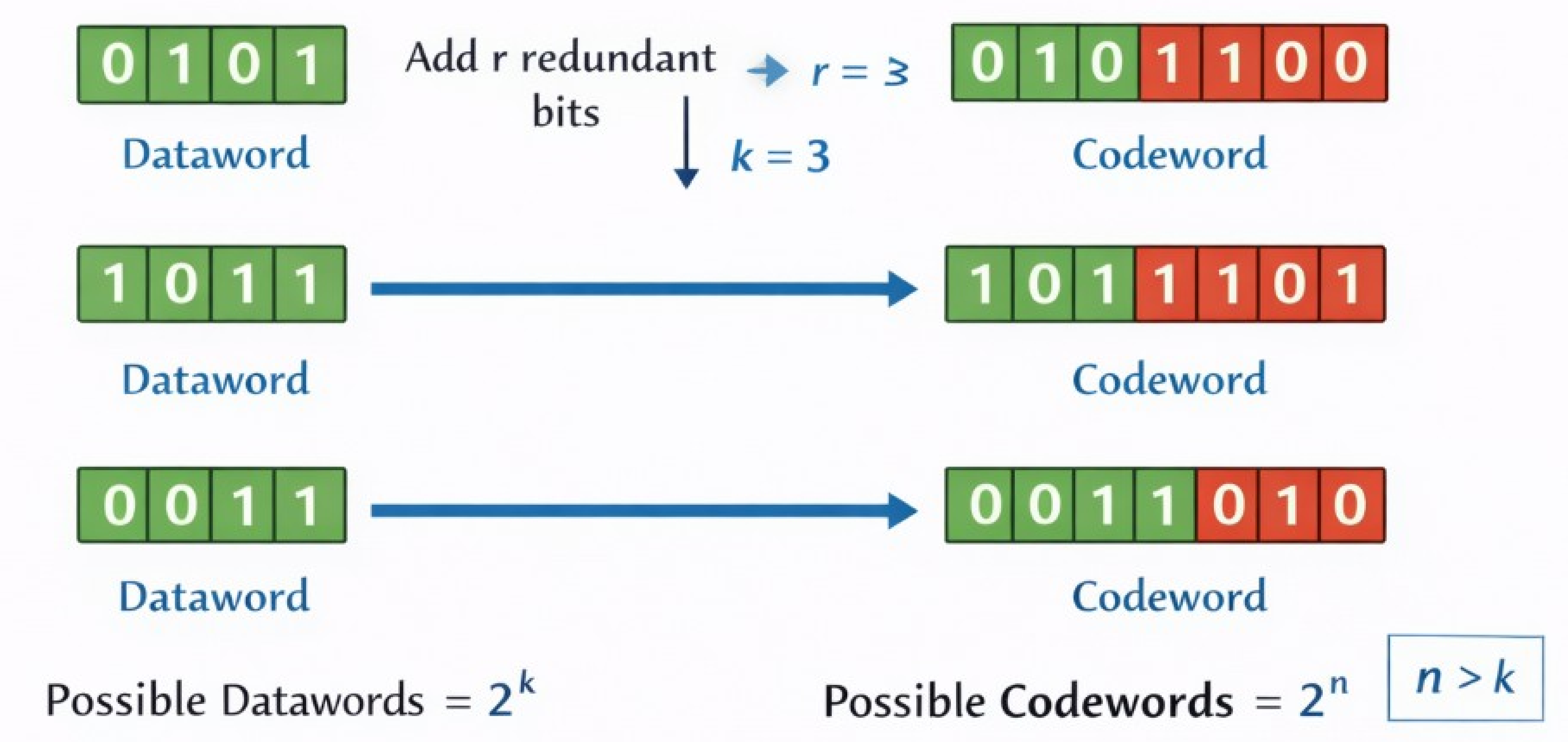
2^k Datawords, each of k bits



Block Coding

Block Coding is an error-control technique in which a block of data bits is replaced by a larger block of coded bits

We add r redundant bits to each block to make the length $n = k + r$.



Error Detection

If the following two conditions are met, the receiver can detect a change in the original codeword.

1. The receiver has(or can find) a list of valid codewords.
2. The original codeword has changed to an invalid one.

If the codeword is corrupted during transmission but the received word still matches a valid codeword, the error remains undetected.

Error Detection

Assume the sender encodes the dataword 01 as 011 and sends it to the receiver. Consider the following cases:

Dataword	Codeword
00	000
01	011
10	101
11	110

Case No.	Sent Codeword	Received Codeword	Error During Transmission	Valid Codeword?	Action at Receiver	Result
1	011	011	No error	Yes	Dataword 01	Correct reception
2	011	111	One bit corrupted	No	Codeword discarded	Error detected
3	011	000	Two bits corrupted	Yes	Dataword 00	Error undetected

Error Correction

Assume the dataword is 01. The sender consults the table (or uses an algorithm) to create the codeword 01011. The codeword is corrupted during transmission, and 01001 is received

Dataword	**Codeword**
00	00000
01	01011
10	10101
11	11110

Codeword No.	Codeword in Table	Received Codeword	Number of Different Bits (Hamming Distance)	Possible Original Codeword?
1	00000	01001	2	No
2	01011	01001	1	Yes
3	10101	01001	4	No
4	11110	01001	3	No

Hamming Distance

The hamming distance between two words is the number of differences between corresponding bits.

E.g. The Hamming distance d(OOO,011) is 2

The Hamming distance d(10101, 11110) is 3

Minimum Hamming Distance

In a set of words, the minimum Hamming distance is the smallest Hamming distance between all possible pairs.

Dataword	Codeword
00	000
01	011
10	101
11	110

Find the minimum Hamming distance of the coding scheme in given Table

Dataword	Codeword
00	00000
01	01011
10	10101
11	11110

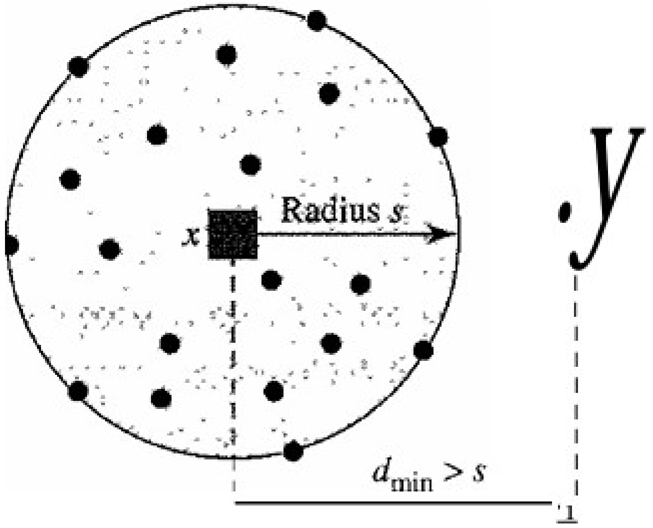
Error Detection

Assume a codeword is transmitted from the sender to the receiver over a noisy channel.

If s bit errors occur during transmission,

then the Hamming distance between the sent codeword and the received codeword is s (because Hamming distance = number of differing bits).

To detect errors, the receiver must be able to recognize that something went wrong. The received codeword must NOT match any valid codeword in the codebook.

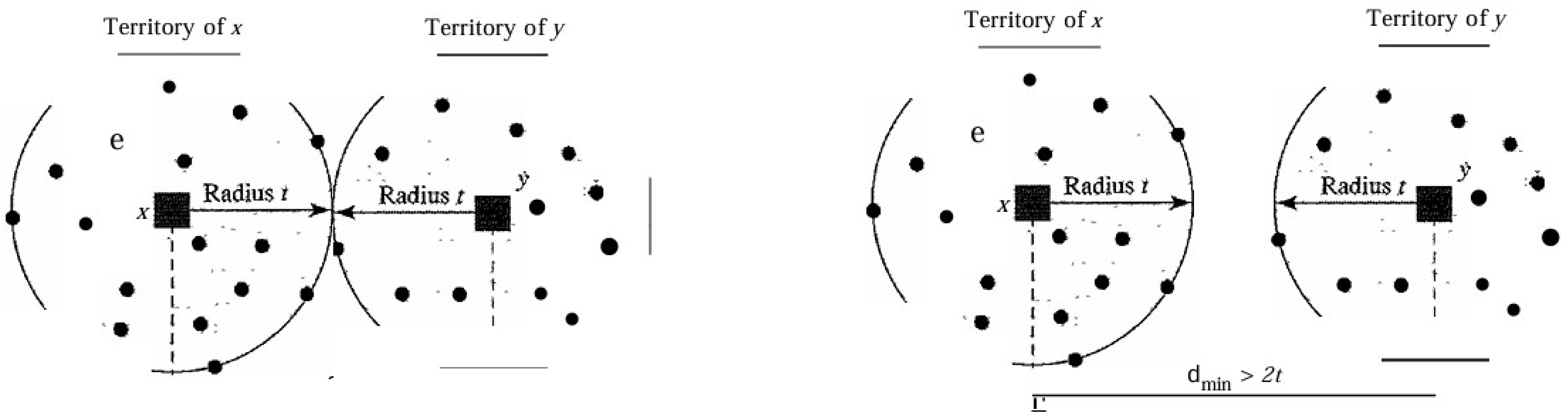


To **detect up to s errors**, the code must have

$$d_{\min} \geq s+1$$

Error Correction

When a received codeword is not a valid codeword, the receiver needs to decide which valid codeword was actually sent.



To guarantee correction of up to t errors in all cases, the minimum Hamming distance in a block code must be $d\_min = 2t + 1$

Linear Block Code

In general block codes, the sender and receiver need a lookup table to map:

- datawords $\longrightarrow$ codewords
- received codewords $\longrightarrow$ nearest valid codewords

As k and n grow, the table size becomes huge.

In a linear block code, the exclusive OR (XOR) of any two valid codewords creates another valid codeword.

Linear Block Code

Minimum Hamming Code

The minimum Hamming distance is the number of Is in the nonzero valid codeword with the smallest number of Is.

the most familiar error-detecting code is the simple parity-check code. In this code, a k-bit dataword is changed to an n-bit codeword where n = k + 1. The extra bit, called the parity bit, is selected to make the total number of Is in the codeword even.

Simple Parity Check

A simple parity-check code is a single-bit error-detecting code in which $n = k + 1$ with $d\_min = 2$

Datawords	*Codewords*	*Datawords*	*Codewords*
0000	00000	1000	10001
0001	00011	1001	10010
0010	00101	1010	10100
0011	00110	1011	10111
0100	01001	1100	11000
0101	01010	1101	11011
0110	01100	1110	11101
0111	01111	1111	11110